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PREDICTIVE RISK MODELING, THE STRUCTURED DECISION MAKING MODEL, AND THE ARCHITECTURE OF CHILD WELFARE DECISION SUPPORT

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KEY POINTS

- A well-designed child welfare system benefits from both Structured Decision Making® (SDM) assessments and predictive risk modeling (PRM).
- PRM represents an advancement in child welfare screening and supervisory oversight, with evidence supporting targeted use cases.
- Every PRM in child welfare is built on systems data (records of what agency staff saw, documented, and decided), which are not an independent or full account of what occurred in families.
- SDM® assessments and PRM most often operate at different points in the decision continuum and draw on different information sources. Where they address the same decision point, they do so through complementary mechanisms: PRM surfacing historical patterns, and SDM assessments structuring practitioner judgment in real time.
- The question for state agencies is not whether to modernize, but how to sequence and integrate tools so that each does what it does best and none is asked to do what it cannot.
- Any tool (algorithmic or structured) informing decisions about children and families must meet clear standards of construct validity, transparency, fairness, and governance before deployment, not after.
- Evident Change is committed to holding all tools, including our own, to these standards, and to supporting states in creating decision architectures that are effective, consistent, fair, and built to last.



INTRODUCTION: A MOMENT OF OPPORTUNITY

In the United States, child welfare agencies are operating under conditions of compounding strain. Referral volumes have risen steadily in recent years, and the workforce charged with responding to them has remained chronically understaffed and undertrained. These conditions are a recipe for significant variability in the quality and consistency of the decisions that determine whether children are safe and families receive appropriate support (Inman & Adams, 2026).

Research has documented for years that two workers assessing the same case can reach meaningfully different conclusions—but not because one is wrong and one is right. Rather, the tools and systems available to them provide insufficient structure for a decision of that complexity and consequence (Bartelink et al., 2015; Rossi et al., 1999). State and federal audits have repeatedly found that risk and safety assessments are completed late, inaccurately, or not at all—conditions that have left children in unsafe situations for longer than necessary and, in the most serious cases, contributed to preventable fatalities and near-fatalities (California State Auditor, 2019; Illinois Office of the Auditor General, 2025; US Department of Health and Human Services, Office of Inspector General, 2024, 2025).

Black and Native American children are investigated at disproportionate rates, removed from their families more frequently, and served by systems whose historical practices have embedded those patterns into the administrative data those same systems now rely upon (Rittenhouse et al., 2024; Whicher et al., 2022). In addition, the technology infrastructure supporting child welfare systems tends to exacerbate worker struggles. Case management systems that should organize and present relevant family history at the point of decision have instead been built and implemented primarily as data repositories, requiring workers to manually reconstruct that history across dozens of screens. In one documented case, a worker would need to navigate 42 screens simply to determine whether an adult had a prior history as a perpetrator (Oregon Secretary of State, 2020).

These are the challenges the federal Administration for Children and Families (ACF) is responding to. ACF's March 2026 brief on predictive risk modeling reflects these needs and advances the conversation (Inman & Adams, 2026). The brief connects explicitly to the administration's broader priorities concerning foster care reduction, caregiver recruitment and retention, and the strategic deployment of federal child welfare resources (Adams, 2025). That policy context matters. The field is not being asked merely to consider a new instrument. It is being asked to align its decision infrastructure with a federal modernization agenda that has implications for how states design, fund, and govern their child welfare systems.

Evident Change approaches this conversation as a nonprofit organization that has spent more than three decades in the careful work of building, validating, and continuously improving the tools practitioners use when they sit across the table from a family in crisis. We understand why the current moment feels urgent, and we share ACF's commitment to pursuing progress. What we bring is a methodological lens: one that asks not whether innovation is desirable, but whether specific innovations meet the standards of evidence, transparency, consistency, and fairness that decisions about children and families demand. That lens applies to every model and tool, including our own.

WHAT PREDICTIVE RISK MODELING DOES WELL

PRM is a class of methods that includes algorithms trained on historical administrative data to generate probabilistic estimates of future outcomes for individuals involved in that system (Drake et al., 2020; Cuccaro-Alamin et al., 2017). What distinguishes PRM from established actuarial risk assessment tools is not the underlying logic, which is familiar to the field, but the scale and automation of the analysis (Cuccaro-Alamin et al., 2017; McNellan et al., 2022). Where traditional actuarial tools draw on a small set of manually scored items, PRM tools can incorporate hundreds of data elements drawn directly from case management systems, update in real time as new information is entered (Drake et al., 2020), and generate consistent classifications across high volumes of cases (Grimon & Mills, 2025; Fitzpatrick et al., 2025). Those methodological characteristics translate into practice benefits, particularly in environments where volume, workforce variability, and information fragmentation present constraints on decision quality. Two use cases have accumulated the most credible evidence.

POPULATION-LEVEL HOTLINE TRIAGE AT SCALE

In high-volume hotline environments, referral volume can outpace even skilled workers' capacity to systematically uncover relevant family history. PRM tools can address this directly. By drawing automatically from existing case management records, they can generate risk-based screening classifications before a screener begins their work, without adding documentation burden to an already strained workforce. Evidence from Colorado's multi-county implementation randomized control study suggests that risk-model-supported screening reduced the time required to make referral decisions (Grimon & Mills, 2025) and is associated with improvements in child safety outcomes and reductions in disparities (Fitzpatrick et al., 2025).

SUPERVISORY OVERSIGHT AND QUALITY ASSURANCE

One of the most promising use cases for PRM is supervisory oversight, rather than frontline screening. When configured to find high-complexity cases for supervisory review, these tools can serve as a meaningful check on workload distribution and case monitoring consistency. The Los Angeles County Department of Children and Family Services (DCFS) pilot, launched in 2021 and focused specifically on supervisory consultation rather than hotline screening, produced improvements in safety outcomes including reductions in fatality and near-fatality measures in pilot offices, with no increase in foster care placement rates or disparities (Los Angeles DCFS, 2022, 2024).

A NOTE ON REAL-TIME UPDATING

Real-time updating is a methodological feature of PRM rather than a distinct use case. PRM tools can update continuously as new information enters the case management system (Drake et al., 2020), meaning that emerging risk can be seen as case conditions change rather than only at fixed assessment intervals.

WHAT PREDICTIVE RISK MODELING CANNOT DO ALONE

Every PRM in child welfare (or any system) is only as good as the data it is trained on. In this field, those data are not a neutral record of what actually happened to children in families. The data are a record of what the system “saw”: what workers chose to document; what agencies deemed important enough to capture; and what decisions professionals made when operating under the same constraints of time, training, and bias that PRM is ostensibly designed to correct.

When a PRM learns from that record, it is learning the patterns of how a particular system has historically processed particular families—not discovering independent truths about maltreatment risk. Those two things are related, but they are not the same. The gap between them is not a data quality issue that better collection practices can fully resolve. It is a problem that is structural to the PRM approach.

The methodological and practical concerns that follow from this fact are broader than any specific model. They are inherent features of building prediction engines on systems data and applying them to decisions about individual families, and any agency considering adoption deserves a clear-eyed account of them.

THE CONSTRUCT VALIDITY PROBLEM

The most fundamental challenge with PRM in child welfare is construct validity: the gap between what the model measures and what child welfare agencies need to know to successfully do their work. This gap originates in the limitations of the data sources that these models rely on.

Child welfare case records document system activity. They capture who came to the system’s attention, what workers chose to document, and what decisions were made, not what actually happened inside a family (Eubanks, 2018; Samant et al., 2021). PRM compounds this problem in a specific way: These models learn patterns from thousands of past cases in the aggregate, then use those patterns to make predictions about one child or family. What a system did historically across many families is being relied upon to tell us what may happen in a particular family. The model is doing something the data were never designed to support—drawing individual-level conclusions from system-level patterns.

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Patterns in how the child welfare system has historically processed or responded to families are not patterns of maltreatment risk independent of system behavior. The gap between these two things is precisely where bias enters: If certain communities have been investigated more frequently or more intensively, the model will learn and reproduce those patterns at scale (Rittenhouse et al., 2024). Crucially, this gap also creates a problem that runs in the opposite direction. Families in communities that have historically had less contact with the child welfare system may carry real risk that simply never appeared in the administrative record. Because the model cannot see what the system never documented, those families are effectively invisible to it—not because they are low risk, but because the system does not know they exist.

The ACF brief acknowledges this critique directly, noting that child welfare data reflect biased historical decisions, and calling for validation against outcomes not solely derived from child welfare processes, such as hospitalization data for maltreatment-related injuries (Inman & Adams, 2026). That is the right standard. Research on the Allegheny Family Screening Tool demonstrates what this kind of validation looks like in practice (Vaithianathan et al., 2020). The question for agencies is simple: Has this tool been validated against something the child welfare system did not itself produce?

TRANSPARENCY AND AUDITABILITY

When a child welfare worker uses an SDM tool to assess a family, every factor that contributed to the final risk classification is visible, documented, and explainable. The worker can point to exactly what information drove the assessment. A supervisor can review it. A family can be told why a decision was made or even see the assessment themselves. A court can examine it. That transparency is not incidental. It is a design principle, and it matters enormously when the decisions being made can result in a child being removed from their home or a family having a case opened for investigation.

Most predictive risk models lack this kind of transparency. The pathway from data to score is often invisible even to the people who built the model. The model may weight certain factors heavily without anyone being able to fully explain why, or identify combinations of variables that predict risk in ways that are statistically defensible but cannot be explained in terms that a worker, family, or reviewing court can examine or challenge. This is sometimes called the “black box” problem, and it is well documented in the literature on algorithmic decision making in public systems (Gerchick et al., 2023). When a family asks why they were flagged, or a caseworker tries to explain a score to a supervisor, or an attorney challenges a decision in court, the answer cannot simply be “the algorithm said so.”

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The inability to explain what drove a risk score is more than a technical inconvenience. It creates legal liability. Families who receive decisions informed by opaque algorithmic scores—decisions that may restrict their access to services, trigger investigation, or result in child removal—have a due-process interest in knowing the basis for those decisions. Agencies that cannot provide that explanation face exposure in administrative proceedings, court challenges, and civil rights litigation (Citron & Pasquale, 2014). It is also a fairness concern and a practice integrity concern that agencies must be prepared to address before (not after) these tools are deployed.

THE FATALITY SAMPLE PROBLEM AND THE LIMITS OF TEXT-BASED ANALYSIS

Some of the most widely discussed PRM tools and analytic projects in child welfare have been developed or validated using child fatality cases as their primary training data (Church & Fairchild, 2017; Drake et al., 2020; Jackson & Marx, 2017). Essentially, the model is built by looking backward at cases where a child died and asking what patterns in the data those cases had in common. The intuition is understandable: If we can identify what fatality cases looked like, we can flag similar cases earlier.

However, this approach carries a considerable methodological problem. Child fatalities, while devastating, represent the extreme outer edge of the maltreatment data distribution. They are statistically rare and may be driven by factors quite different from those that predict maltreatment risk in the broader population of families who come to the system's attention.

Building a systemwide screening tool on predictive scores derived from fatality cases can set off alarms for the wrong reasons in the wrong cases. Moreover, it focuses an entire system's architecture on factors shared by a miniscule proportion of families. An overwhelming majority of families who enter the system need only a low level of intervention. Federal data show that roughly half of referrals are screened out before investigation and that substantiated cases represent a small fraction of total system contact (Children's Bureau, 2024; Drake et al., 2020; Inman & Adams, 2026). An architecture calibrated to the rarest and most extreme cases may misidentify the needs of the majority of families in the system, directing resources away from where they would do the most good.

A related and increasingly common approach involves using artificial intelligence, specifically large language models (LLMs) or natural language processing (NLP) tools, to analyze the written notes that caseworkers enter into child welfare records. These are the same technologies that power tools like ChatGPT: AI systems trained to read, interpret, and draw conclusions from text. The appeal in child welfare is obvious; caseworker narratives contain rich contextual information that structured data fields cannot capture.

But applying these tools to CPS case notes reintroduces the systems data problem in a new manner. The notes being analyzed were written by workers who were themselves subject to the same biases, time pressures, and documentation norms that shape all child welfare records. When an AI reads those notes to identify risk factors, it is not reading an objective account of a family. Instead, it is reading a human worker's interpretation of a family, filtered through the lens of a system with a well-documented history of disparate practice. Published research confirms this concern directly: Studies applying NLP to child welfare case notes have found meaningful bias in the outputs, raising questions about whether this approach can produce fair results at scale (Field et al., 2023).

For a child welfare leader, the practical consequence is direct: Deploying an LLM or NLP tool to analyze case notes risks automating and amplifying the disparities the agency is working to reduce, and doing so at a speed and scale that makes those patterns much harder to detect and correct. Without the framework of a structured tool that requires workers to gather and assess contextual information directly with families, agencies risk making high-stakes decisions at a distance from the very people those decisions affect—squandering resources, undermining fairness, and in the most serious cases, causing harm.

MODEL DRIFT, GOVERNANCE FAILURE, AND THE LIMITS OF ACCOUNTABILITY

Even well-validated models degrade over time. As agency practices, workforce composition, and community demographics change, the historical patterns on which a model was trained become less representative of current conditions (Whicher et al., 2022). This degradation is not always obvious in aggregate performance metrics. A model can appear to be functioning while quietly performing worse for specific subpopulations or case types. Without routine recalibration, subpopulation performance monitoring, and structured feedback loops for continued quality improvement (requirements the ACF brief endorses), PRM tools will deliver decreasing accuracy over time in ways that frontline users have no way to detect or correct.

The consequences of inadequate governance are not hypothetical. Illinois was among the first states to adopt a PRM tool in child welfare, implementing the Eckerd Connects Rapid Safety Feedback tool as part of a broader effort to modernize child welfare decision making following a series of high-profile child deaths. The results were instructive in ways the program's architects did not intend. The tool simultaneously over-flagged thousands of families (assigning a 90% or greater probability of death or serious injury to more than 4,100 children, and a 100% probability to 369 children under age nine) while failing to flag the actual fatalities that occurred (Jackson & Marx, 2017). Illinois Department of Children and Family Services Director Beverly Walker terminated the program, stating that the tool "didn't seem to be predicting much" (Jackson & Marx, 2017).

Compounding these performance failures were critical governance problems. The algorithm was entirely proprietary, and neither workers nor families had any ability to understand or challenge the generated scores.

The Illinois experience is not an argument against PRM adoption. It is a detailed illustration of what happens when tools are deployed without the governance infrastructure (independent validation, ongoing monitoring, and clear accountability mechanisms) that responsible adoption requires.

SDM ASSESSMENTS: DIFFERENT TOOLS FOR A DIFFERENT NEED

A recurring feature of the current policy conversation is a conflation that, left unaddressed, is likely to produce implementation problems. When agencies treat SDM tools and PRM as interchangeable alternatives competing for the same role, the predictable results are gaps in practice consistency, misallocated resources, and decision-making architectures that lack the structured, accountable engagement that protects both children and the agencies serving them (Drake et al., 2020; Yu & Rose, 2025). Research on decision-making variability in child welfare consistently demonstrates that inconsistent, poorly structured practice is associated with worse case outcomes (Bartelink et al., 2015; Rossi et al., 1999). A system that conflates these two tools and their use cases risks recreating these conditions at scale with consequences for the consistency, fairness, and defensibility of decisions made about children and families.

A system that conflates PRM and SDM and their use cases risks consequences for the consistency, fairness, and defensibility of decisions made about children and families.

SDM tools were intentionally designed to structure professional judgment at the moment a trained practitioner is in direct contact with a family. They are built on actuarial research validated against recurrence outcomes (Baird et al., 1999), with each item carrying a documented evidentiary basis (Cuccaro-Alamin et al., 2017). They are transparent and deterministic, meaning that the same information produces the same classification every time; and that classification can be fully explained to the worker, supervisor, family, or reviewing court. Systematic reviews of the evidence base for structured risk tools confirm that actuarially grounded instruments outperform unstructured clinical judgment in predictive accuracy (McNellan et al., 2022).

Most importantly, SDM tools structure what happens when a practitioner engages with a family, including what information is gathered, how risk is classified, what safety decisions are made, and what services are appropriate. An algorithmic score operating retrospectively on administrative records cannot substitute for that practitioner–family encounter or the structured documentation of what was learned in it. Critically, that encounter is also the point at which information that exists nowhere in any administrative record becomes available: what the family has already tried, what they trust, what they fear, and what they need. Predictive risk models cannot surface that knowledge. Only a trained practitioner in direct engagement with a family can gather it, and only a structured tool can ensure that what is gathered is documented, consistent, and accountable.

The ACF brief raises legitimate concerns about inter-rater reliability in manual checklist tools (Inman & Adams, 2026), and those concerns have a basis in the literature (Bartelink et al., 2015; Rossi et al., 1999). Research has documented considerable inconsistency in child protection decision making across workers assessing similar cases. The solution to inconsistent application of structured tools is better training, fidelity monitoring, and instrument maintenance—not replacing structured tools with algorithmic scores that provide no mechanism for professional calibration.

A FRAMEWORK FOR COMPLEMENTARITY

The following table maps PRM and the SDM model along key attributes of child welfare decision making. Taken together, these elements reveal not a competition but a continuum of guidance and support.

Attribute of Decision Making	Predictive Risk Modeling (PRM)	Structured Decision Making (SDM) Model
Primary Purpose	Population-level triage; identify which cases warrant closer attention across high referral volumes	Structure professional judgment at defined decision points throughout a case
Data Source	Administrative records, applied algorithmically and automatically	Administrative data used to build and validate actuarial tools; practitioner observation applied at the case level

Attribute of Decision Making	Predictive Risk Modeling (PRM)	Structured Decision Making (SDM) Model
Model Development	Machine learning algorithms trained on historical patterns; developed and deployed algorithmically	Actuarial research validated locally against recurrence outcomes; consensus-driven decision trees collaboratively developed with practitioners and agency stakeholders
Output	Probability score or risk flag	Structured classification with defined, transparent decision pathways
Transparency	Often opaque; pathways from data to score not fully explainable	Fully deterministic and auditable; every factor, weight, and pathway documented and explainable
Worker Role	Receives algorithmic signal to inform inquiry	Exercises structured, documented professional judgment within a validated framework
Practitioner Interaction	Worker receives an algorithmic score or flag via a dashboard or alert; the scoring pathway is inside the algorithm and not directly accessible	Worker interacts with a structured instrument (a defined set of items and weights) that they apply, document, and can explain
Fairness Mechanism	Requires ongoing post-deployment monitoring for subpopulation bias	Bias addressed through local validation, tool development process, and structured override documentation
Core Question	Where should we direct attention? Which cases warrant closer review?	What are we seeing in this family? What level of risk and safety concern exists, and what should happen next?

The practical implication is straightforward: Well-designed child welfare decision architecture can use PRM to manage population-level signals identifying which cases warrant elevated attention, where supervisory consultation is warranted, and where low-risk referrals can be safely handled with lighter-touch intervention. SDM tools then structure what happens when a practitioner engages with the family. These are sequential and complementary functions, not competing ones.

A SCORE IS NOT THE SAME AS A TOOL

One of the most consequential differences between PRM and the SDM model is what the worker interacts with at the point of decision. In most PRM implementations, the algorithm runs continuously in the background and conveys its output as a score, a flag, or an alert: in essence, a piece of information. The pathway from data to score remains inside the algorithm, inaccessible to the practitioner basing decisions on it.

The SDM model works differently. Actuarial research happens upstream (in the development and local validation process), and what the practitioner receives is the resulting tool that guides the case process. This tool is a defined set of items, transparent weights, and a classification they can explain to a supervisor, a court, or a family. This distinction matters for training and for accountability. States considering PRM adoption should ask a direct question of any vendor: Are we getting a tool, or are we getting a score? The answer should shape how the system will function in practice.

A familiar example illustrates the difference between a score and a tool. An entertainment streaming service algorithm analyzes your viewing history and serves up recommendations—some that feel exactly right, and others that are perplexing. The algorithm cannot know that you are not in the mood for a thriller, that the film stars someone you dislike, or that your children are sitting next to you on the couch. The score is useful input for your decision about what to watch. But the decision requires something the algorithm does not have: direct knowledge and experience of the context of the moment, and a framework for applying your own judgment to it. Using both the recommendations and your own framework for decision making, you can arrive at a decision that fits your needs exactly.

Well-designed child welfare decision architecture can use PRM to manage population-level signals; SDM tools then structure what happens when a practitioner engages with the family. These are sequential and complementary functions, not competing ones.

A PRM score works similarly. It can recognize patterns in historical data that a worker might not have time to find on their own. What it cannot do is sit across the table from a family, observe what is present, and apply a structured framework to document what it learns there. This is why a score, however accurate in the aggregate, is not a substitute for a tool in the hands of a trained practitioner.

This sequencing—using algorithmic output to direct attention, followed by structured professional judgment applied directly with a family—has emerging empirical support. Evidence from Colorado and Los Angeles suggests that the greatest gains in child safety came not from replacing structured practice with algorithmic

scores, but from using predictive tools to direct supervisory attention and consultation—functions that then interfaced with structured casework practice (Grimon & Mills, 2025; Inman & Adams, 2026).

Most importantly, SDM tools structure what happens when a practitioner engages with a family, based on what information is gathered, how risk is classified, what safety decisions are made, and what services are appropriate. As noted earlier, an algorithmic score operating retrospectively on administrative records cannot substitute for the practitioner–family encounter or the structured documentation of what was learned in it.

WHAT STATES AND AGENCIES SHOULD CONSIDER BEFORE ADOPTING ANY TOOL

Whether evaluating a PRM tool, an SDM tool, or any other decision-support model or tool, Evident Change recommends that state agencies require clear, documented evidence that the following considerations have been addressed.

- **Construct Validity.** The validation standard for any decision-support tool should match the scope of the claim that tool is making, and these claims differ meaningfully across tool types. Actuarial risk assessment instruments used in the SDM model are designed to predict the likelihood of child welfare system contact and are appropriately validated against that outcome (Baird & Wagner, 2000; Coohey et al., 2013; McNellan et al., 2022; Shlonsky & Wagner, 2005). That is a bounded and transparent claim: Given these factors, as assessed by a practitioner in direct contact with a family, the likelihood of future system involvement is elevated.

Developers of PRM-based tools frequently make a broader and fundamentally different claim: that an algorithm operating on administrative records can predict harm, serious injury, or death, independent of practitioner contact. To justify this claim, the appropriate validation standard is a test of performance against indicators *outside* the child welfare system (such as emergency room visits or documented injury hospitalizations). Validating a harm-prediction model solely against future reports or removals from the system’s administrative data would confirm only that the model has learned what the system does—not whether children are actually safer (Vaithianathan et al., 2020).

- **Transparency.** Can the classification produced by this tool be fully explained to a practitioner, supervisor, family, or reviewing court? Is the pathway from input to output traceable and documented? (Gerchick et al., 2023)
- **Tool or Score.** Agencies should clarify how staff will interact with the instrument or tool at each point of decision making. Will workers apply a structured instrument with defined items, documented weights, and an explainable output, or will they receive a score or algorithmic flag generated from background processing? Understanding this distinction is important for planning workforce training, establishing documentation protocols, and ensuring that the agency can account for what factors and processes informed specific case decisions (Drake et al., 2020; Gerchick et al., 2023).
- **Fairness.** Does the tool perform with equal accuracy across the populations it will be used to assess? Disparities in false positive and false negative rates across racial, ethnic, and socioeconomic groups

are not just a fairness concern; they are a performance concern and a legal liability. Agencies should require subgroup-level performance data before deployment and compare them against current practice outcomes for those same groups (Rittenhouse et al., 2024; Whicher et al., 2022).

- **Practitioner Integration.** How is this tool integrated into existing practice protocols? What training is required? What happens when a practitioner's direct assessment conflicts with an algorithmic score? Are overrides documented and reviewed? (Drake et al., 2020)
- **Governance and Sustainability.** What is the plan for ongoing monitoring, recalibration, and performance review? Who is responsible for identifying and responding to model drift or subpopulation disparities, and at what frequency? (Whicher et al., 2022)

These questions apply equally to PRM and SDM tools. Evident Change has maintained rigorous validation and recalibration processes for SDM instruments across jurisdictions for decades, and we hold our own tools to the standards we describe here.

WHAT WE OWE CHILDREN AND FAMILIES

Evident Change is committed to improving outcomes for children and families. That commitment requires ongoing evaluation of every instrument, including our own. We believe the field is at an inflection point and *getting this moment right matters*. States that adopt PRM tools without robust governance, subgroup performance monitoring, and practitioner integration frameworks risk replicating and amplifying the disparities and inconsistencies that motivated the search for better tools in the first place (Samant et al.,

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2021; Whicher et al., 2022). States that dismiss algorithmic approaches entirely will miss opportunities to improve screening consistency, supervisory oversight, and resource targeting.

The path forward is neither uncritical adoption nor reflexive resistance. It is principled integration: bringing the best of what automated pattern recognition and structured professional judgment

each offer into a coherent decision architecture that is transparent, consistent, fair, and accountable to the children and families it serves. The commitment to getting this right is ultimately a commitment to them.

Children and families are not data points in this conversation. They are the purpose of the conversation. Every tool we develop and deploy, every policy or standard we set, and every framework we help build should be measurable against a simple standard: making it more likely that a family in crisis receives an accurate, fair, and consistent response anchored in what they are truly experiencing, not simply a continuation of a pattern the system has seen before.

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GLOSSARY OF KEY TERMS

Actuarial Tool: A decision-support instrument that uses statistically derived weights, based on research about what factors are most predictive of a specific outcome, to produce a structured risk classification. Actuarial tools are transparent: Every item and its weight can be examined, questioned, and explained.

Construct Validity: The degree to which a tool measures what it claims to measure. A tool with poor construct validity may produce scores that reflect something other than the outcome it was designed to predict.

False Positive / False Negative: A false positive occurs when a tool flags a case as requiring a higher level of response that does not. A false negative occurs when a tool fails to flag a case that does. In child welfare, both errors have important consequences: False positives expose families to unnecessary investigation and consume limited agency resources; false negatives can leave children without timely intervention.

What matters as much as the overall error rate is how errors are distributed across populations. When false positive rates are significantly higher for some groups than others (a pattern documented in the research literature), the tool is not performing equally for all families, regardless of how it performs in the aggregate. That gap is both a performance problem and an accountability problem that agencies must examine before implementation and deployment.

Inter-Rater Reliability: The degree to which two different workers, assessing the same case with the same tool, reach the same conclusion. Low inter-rater reliability means that outcomes for families depend heavily on which worker happens to be assigned to their case rather than on the facts of the situation.

Large Language Model (LLM) / Natural Language Processing (NLP): Artificial intelligence technologies designed to read, interpret, and draw conclusions from written text. LLMs are the technology underlying tools like ChatGPT. NLP is a related approach used to analyze patterns in large volumes of text (such as case notes) to identify themes or risk indicators. When applied to child welfare records, these tools inherit the limitations and biases embedded in the text they analyze.

Model Drift: The degradation in a predictive model's accuracy over time as the conditions it was trained on change. A model trained on data from five years ago may perform poorly today if agency practices, workforce composition, or community demographics have shifted. Without regular recalibration, drift can go undetected while the model continues to generate scores that appear authoritative.

Predictive Risk Model (PRM): A class of algorithms that analyze patterns in historical administrative data to generate a probability estimate that a specific future outcome (such as subsequent maltreatment or system involvement) will occur for a given child or family. PRMs are distinct from structured tools in that they operate automatically on existing records rather than requiring a practitioner to complete an assessment.

Structured Decision Making (SDM) Model: A framework of evidence-based tools designed to bring consistency, transparency, and accountability to the decisions child welfare workers make at key points in a case, including safety assessment, risk assessment, family strengths and needs assessment, and reunification assessment. SDM tools support professional judgment; they do not replace it.

Training Data: The historical records used to build a predictive model. A model “learns” by identifying patterns in its training data and using those patterns to make predictions about new cases. The quality, completeness, and representativeness of training data directly determine what a model can and cannot reliably predict.

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